

- 2 The perimeter of a rectangle is 56 cm. If the length of the rectangle is three times its width, determine the dimensions by setting up and solving two simultaneous equations.
- 3 The sum of the ages of two children is 24 and the difference between their ages is 8. If Nikki is the older sister of Travis, determine their ages by setting up and solving a pair of simultaneous equations.
- 4 Cam is 3 years older than Lara. If their combined age is 63, determine their ages by solving an appropriate pair of equations.
- 5 Alex buys 4 bolts and 6 washers for \$2.20 and Holly spends \$1.80 on 3 bolts and 5 washers at the same local hardware store. Determine the costs of a bolt and a washer.
- 6 It costs \$3 for children and \$7 for adults to attend a basketball game. If 5000 people attended the game and the total takings at the door was \$25 000, determine the number of children and the number of adults that attended the game.
- 7 A vanilla thickshake is \$2 more than a fruit juice. If 3 vanilla thickshakes and 5 fruit juices cost \$30, determine their individual prices.
- 8 A paddock contains both ducks and sheep. There are a total of 42 heads and 96 feet in the paddock. How many ducks and how many sheep are in the paddock?



Revision

- 4 Solve these linear equations.

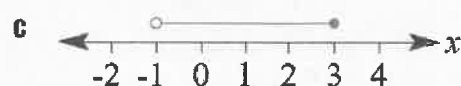
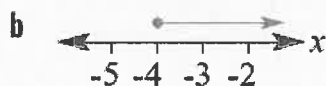
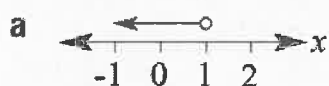
a $3 - 2x = 9$

b $3(2x + 1) = 7 - 2(x + 5)$

c $\frac{5x - 9}{4} = -2$

d $\frac{2x - 1}{3} = \frac{x + 2}{4}$

- 5 Write as an inequality.



- 6 Solve the following inequalities.

a $4x - 3 > 17$

b $3x + 2 \leq 4(x - 2)$

c $1 - \frac{x}{3} < 2$

d $-2x \geq -4(1 - 3x)$

8 Sketch graphs of the following linear relations, labelling the points where the graph cuts the axes.

a $y = 3x - 9$

b $y = 5 - 2x$

c $y = 3$

d $x = 5$

e $y = 2x$

f $y = -5x$

g $x + 2y = 8$

h $3x + 8y = 24$

12 Determine the equation of the line that is:

a parallel to the line $y = 3x + 8$ and passes through the point (2, 4)

b parallel to the line with equation $y = 4$ and passes through the point (3, -1)

c perpendicular to the line $y = 2x - 4$ and has a y-intercept of (0, 5)

d perpendicular to the line with equation $x + 3y = 5$ and passes through the point (2, 5).

14 Solve the following simultaneous equations using the substitution method.

a $y = 5x + 14$

b $3x - 2y = 18$

$y = 2x + 5$

$y = 2x - 5$

15 Solve these simultaneous equations by elimination.

a $3x + 2y = -11$

b $2y - 5x = 4$

$2x - y = -5$

$3y - 2x = 6$

16 At the movies Jodie buys 3 regular popcorns and 5 small drinks for her friends at a cost of \$24.50. Her friend Renee buys 4 regular popcorns and 3 small drinks for her friends at a cost of \$23.50. Find the individual costs of a regular popcorn and a small drink.

Answers

2 7 cm $\times$ 21 cm

3 Nikki is 16, Travis is 8.

4 Cam is 33, Lara is 30.

5 Bolts cost \$0.10, washers cost \$0.30.

6 There were 2500 adults and 2500 children.

7 Thickshakes cost \$5, juices cost \$3.

8 There are 38 ducks and 8 sheep.

4 a $x = -3$ b $x = -\frac{3}{4}$ c $x = \frac{1}{5}$ d $x = 2$

5 a $x < 1$ b $x \geq -4$ c $-1 < x \leq 3$

6 a $x > 5$ b $x \geq 10$ c $x > -3$ d $x \leq \frac{2}{7}$

12 a $y = 3x - 2$ b $y = -1$ c $y = -\frac{1}{2}x + 5$

d $y = 3x - 1$

13 a $x = 7$ b $b = -8$ c $c = 0$ or 4

14 a (-3, -1) b (-8, -21)

15 a (-3, -1) b (0, 2)

16 A regular popcorn costs \$4 and a small drink costs \$2.50.